



ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG QUỐC TẾ
VNU-INTERNATIONAL SCHOOL

MODULE SYLLABUS

DATABASE

- 1. Module code:** INS2080 (Compulsory ☒ / Elective ☐)
- 2. Credit(s):** 3
- 3. Prerequisite(s):** INS1073
- 4. Teaching language(s):** English
- 5. Program:** Undergraduate, Major: Management Information System
- Faculty in charge:** Faculty of Applied Sciences
- 6. Lecturers' information:**

N o	Name	Title/Qualification	Email	Cell phone
1.	Tran Thi Ngan	Assoc. Prof. Dr.	ngantt@vnuis.edu.vn	0989040454
2.	Michael Omar	Master	Omar2@fe.edu.vn	0326189705

7. Module objectives: (*general objectives, specific objectives: knowledge, skills, attitude*):

The main aim of this module is to provide students with database design & implementation knowledge & methods to develop relational databases.

For knowledge, students should be able to identify and implement database models; contrast and compare a variety of database technologies; cite relevant legislation relating to issues such as Data Protection, Privacy and Copyright; discuss and select a suitable

data management system for managing and organization's data; identify and implement database models; examine and design a data model for an organization; select and construct different methods of organizing files to ensure efficient storage and retrieval; program and implement database solutions using basic SQL statement.

For skills and attitude, students can develop further on the capacity of thinking and deducing logically and systematically, computing skills; understand the importance and be aware of applying mathematics to real-world problems.

8. Learning outcomes:

By successful in INS2037, students will be able to:

8.1. Knowledge

COURSE LEARNING OUTCOMES (CLO)	DETAILS	PROGRAM LEARNING OUTCOMES (PLO)
CLO 1	UNDERSTAND the difference between a relational database and a flat file	PLO 3
CLO 2	REMEMBERING and UNDERSTANDING database models;	PLO 3
CLO 3	UNDERSTANDING the relational data model	PLO 3
CLO 4	REMEMBERING the main characteristics of entity relationship components UNDERSTANDING real-world database design often requires the reconciliation of conflicting goals	PLO 3
CLO 5	APPLYING flexible solutions for special data-modeling cases and APPLYING normalization rules to evaluate and correct table structures	PLO 3
CLO 6	UNDERSTANDING the key principles in crafting a SELECT query and APPLYING SQL to create a table	PLO 3

	manually	
CLO 7	UNDERSTANDING the role of database design as the foundation of a successful information system and UNDERSTANDING the database transaction management process	PLO 3
CLO 8	UNDERSTANDING how a DBMS processes SQL queries in each of its three phases and UNDERSTANDING the purpose and function of distributed database management systems (DDBMSs)	PLO 3
CLO9	UNDERSTANDING the role of business intelligence in providing comprehensive business decision support and UNDERSTANDING the role of Big Data in modern business	PLO 3

8.2. Skills

* Professional skills

CLO	DETAILS	PLO fulfillment
CLO 10	UNDERSTAND the importance and BE AWARE of designing databases in helping company store, analyze, and extracting information;	PLO 6

* Soft Skills

CLO	DETAILS	PLO fulfillment
CLO 11	IMPROVE further on the ability to think and reason logically and systematically;	PLO 10
CLO 12	BUILD up database vocabulary in English; ENHANCE presentation skills in English.	PLO 9, 10

8.3. Level of Autonomy and Responsibility

CLO	DETAILS	PLO fulfillment
CLO 13	Students have good moral qualities, appropriate personal qualities such as willingness to face difficulties and take risks, perseverance, flexibility, confidence, hard work, enthusiasm, passion, self-control, integrity, criticism, desire for improvement and innovation, creativity, responsibility and initiative in work.	PLO 12, 14

9. Assessment methods

No .	Assessments (items and methods)	Proportion	Description	CLO fulfillment	PLO fulfillment
1	Regular assessment	10%	Homework + class participation	CLO 1-13	PLO 12, 14
2	Midterm assessment	30%	Two-hour exam	CLO 1-13	PLO 3, 6, 9, 10
3	Final exam	60%	Group project	CLO 1-13	PLO 3, 6, 9, 10
	Total	100%			

Scoring method:

- Module evaluation scores are graded on a 10-point scale (from 0 to 10), rounded to 0.5
- The module score is the sum of two mid-term and final exam scores multiplied by the respective proportion

10. Required textbook(s) and recommended books

10.1. Required textbook(s):

- [1] Coronel C., Morris S. (2019) - Database Systems Design, Implementation, &

Management, 13th edition.

10.2. Recommended text(s):

[1] Elvis Foster, Shripad Godbole (2022) - Database Systems_ A Pragmatic Approach-CRC Press.

11. Module description (summarizes about 120 words)

This course introduces underlying the design and implementation of databases and database management systems. It will cover the theoretical and practical languages that have been developed for relational databases, their implementation and optimization.

12. Course schedule

- + Concepts and techniques are introduced in a weekly three-hour active-learning lecture.
- + Classroom lectures combine with discussion and class assignments, homework;
- + Students present the solution of exercises on the board.

No.	Content	Forms of teaching			Total	Teaching methods	Reference	CLO fulfillment
		Theory teaching	Practice	Self-study				
(1)	(2)	(3)	(4)	(5)	(6) = (3) + (4)	(7)	(1)	(2)
1	Database Systems	2	2	6	3	Lecture, discussion	Chapter 1	1
2	Data Models	2	2	6	3	Lecture, discussion	Chapter 2	2
3	The relational Database Mode	2	2	6	3	Lecture, discussion	Chapter 3	3

No.	Content	Forms of teaching			Total	Teaching methods	Reference	CLO fulfillment
						n		
4	Entity Relationship (ER) Modeling	2	2	6	3	Lecture, discussion	Chapter 4	4
5	Advanced Data Modeling	2	2	6	3	Lecture, discussion	Chapter 5	5
6	Normalization of Database Tables	2	2	6	3	Lecture, discussion	Chapter 6	5
7	Introduction to Structured Query Language (SQL)	2	2	6	3	Lecture, discussion	Chapter 7	6
8	Review & Mid-term	1	4	5	3	Lecture, discussion		1-6
9	Advanced SQL	2	2	6	3	Lecture, discussion	Chapter 8	7
10	Transaction Management and Concurrency Control	2	2	6	3	Lecture, discussion	Chapter 10	7
11	Database Performance Tuning and Query Optimization	2	2	6	3	Lecture, discussion	Chapter 11	8
12	Distributed Database	2	2	6	3	Lecture, discussion	Chapter	8

No.	Content	Forms of teaching			Total	Teaching methods	Reference	CLO fulfillment
	Management Systems					n	12	
13	Business Intelligence and Data Warehouses	2	2	6	3	Lecture, discussion	Chapter 13	9
14	Big Data and NoSQL	2	2	6	3	Lecture, discussion	Chapter 14	9
15	Review	3	0	7	3	Lecture, discussion	Chapter 1-15	1-9
	Total	30	30	90	45			1-13

Hanoi,2023

Approved by

Compiler

Note: Credit hours is a measure of a student's study time, classified into categories according to teaching methods and determined as follows:

a) One theoretical credit hour is equal to 01 theory period.

b) One practical credit hour equals 2-3 practice or discussion periods; 3-6 practice periods at the facility; 3-4 periods for essays, exercise, major assignments or projects, graduation theses.

c) One credit hour of compulsory self-study is equal to 3 compulsory self-study periods and is assessed. Self-study credits are no